DATABASES

PostgreSQL Advanced Visual Guide

JSONB, extensions, partitioning, full-text search -- the database that does everything.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

JSONB: Documents in a Row

Schema-flexible columns with real indexing.

```
CREATE TABLE products (id serial PRIMARY KEY,
  attrs jsonb);
-- query inside the document:
SELECT * FROM products
WHERE attrs @> '{"color": "red"}';
-- index it: CREATE INDEX ON products USING gin (attrs);
```

- Operators: -> (field), ->> (field as text), @> (contains), ? (has key).
- GIN indexes make containment queries fast -- jsonb without GIN is a table scan.
- Rule: JSONB for flexible payloads, real columns for fields you filter/join on.

STEP 2/6

Extensions

CREATE EXTENSION turns Postgres into something new.

PostGIS

geometry/geography types -- the serious GIS database

pgvector

vector type + HNSW/IVFFlat -- embeddings search in SQL

pg_trgm

trigram similarity -- fuzzy text matching

uuid-ossp / pgcrypto

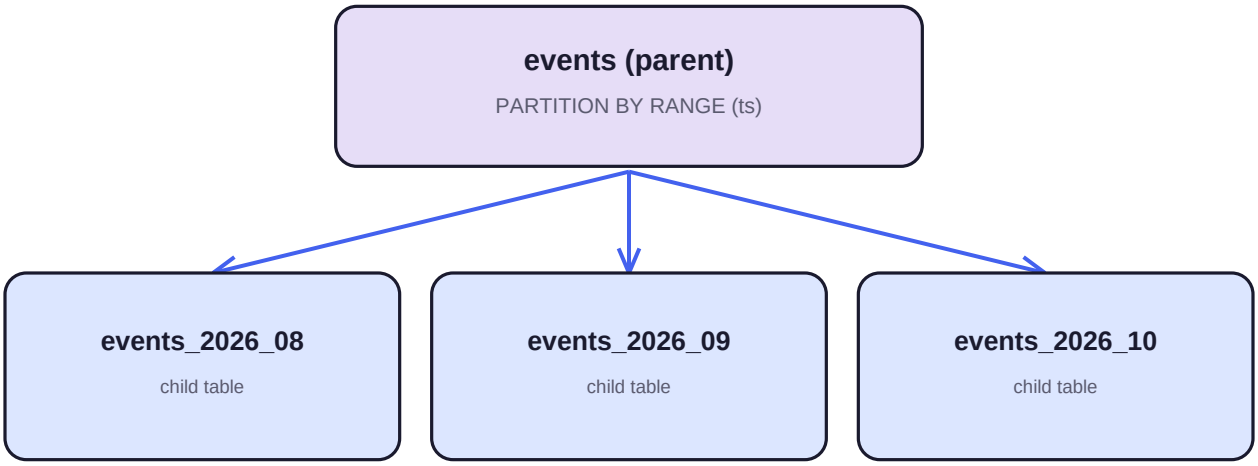
UUIDs, hashing, encryption primitives

Managed clouds allowlist extensions -- verify PostGIS/pgvector availability before designing around them.

STEP 3/6

Table Partitioning

One logical table, many physical children.

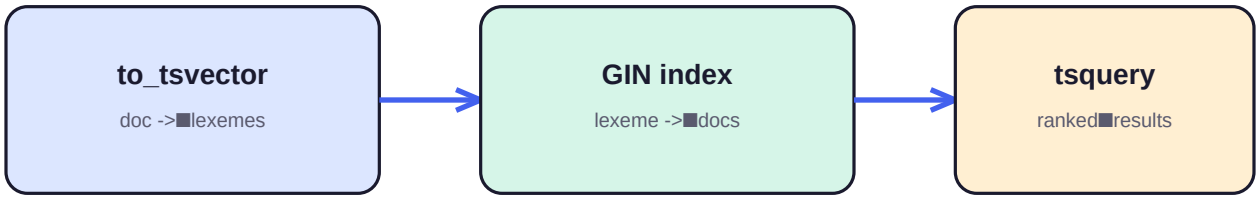


- Partition pruning: WHERE ts >= '2026-10-01' touches only matching children.
- Drop old partitions in milliseconds instead of DELETEing billions of rows.
- Range (time), list (region), hash (even spread) -- pick by query pattern.

STEP 4/6

Full-Text Search

Real search without an external engine -- for moderate scale.




```
SELECT title FROM articles
WHERE to_tsvector('english', body) @@ to_tsquery('postgres & search')
ORDER BY ts_rank(to_tsvector('english', body),
                 to_tsquery('postgres & search')) DESC;
```

Handles stemming, stop words, ranking. Outgrown it? Replicate to OpenSearch/Typesense.

STEP 5/6

Performance Toolkit

Built-in instruments for slow queries.

EXPLAIN (ANALYZE, BUFFERS)

the actual plan + real timings -- always start here

pg_stat_statements

which queries burn the most total time -- the hit list

autovacuum

reclaims dead tuples; never disable it, tune it

pg_stat_user_tables

seq scans vs index scans per table -- missing-index radar

STEP ✓/✓

Key Takeaways

PostgreSQL Advanced -- Remember

- JSONB + GIN gives document flexibility inside a relational engine -- index what you query.
- Extensions (PostGIS, pgvector, pg_trgm) turn Postgres into GIS, vector, and fuzzy-search stores.
- Partition large tables by time/range: pruning speeds queries, dropping beats deleting.
- Full-text search with tsvector/tsquery covers moderate scale before you need a search cluster.
- EXPLAIN ANALYZE, pg_stat_statements, and autovacuum are the core performance loop.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev